GEOMETRY QUEST · WORLD 15

Pythagoras & √2

ANSWER SHEET · for the grown-up

Answers are shown in green. “Check” means look at the child’s drawing or measuring: allow ±2° or ±2 mm.

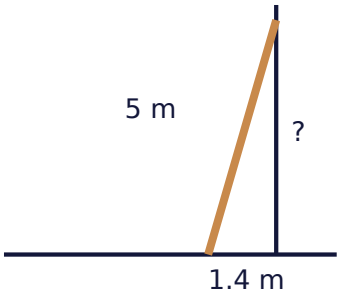
1 Find the hypotenuse (longest side) of each right-angled triangle. $a^2 + b^2 = c^2$.

3 and 4	5 and 12	8 and 15	9 and 12
5	**13**	**17**	**15**

2 Find the missing shorter side.

hypotenuse 10, side 6	hypotenuse 13, side 5	hypotenuse 25, side 7
8	**12**	**24**

3 A 5 m ladder leans on a wall. Its foot is 1.4 m from the wall. How high up the wall does it reach?



4.8 m

√(25 − 1.96) = √23.04 = 4.8

4 Is a triangle with sides 7, 24 and 25 right-angled? Show your working.

$7^2 + 24^2 = 49 + 576 = 625 = 25^2$. Yes, it is right-angled.

5 Paper investigation: measure a sheet of A4. Divide the long side by the short side. Fold it in half and do the same for the new A5 size.

A4: **297 ÷ 210 ≈ 1.41** **A5:** **210 ÷ 148 ≈ 1.42**

Both are about √2 = 1.414…, so the shape stays the same.

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